

SUNIX SDC Expansion Board

SocketCAN

User Guideline

Driver Ver. 2.1.1.0

Doc Ver. 2.1

2025 March



SUNIX CO., Ltd.

Tel: +886-2-8913-1987

Fax: +886-2-8913-1986

<http://www.sunix.com>

info@sunix.com.tw

Index

1.	SocketCAN	3
1.1.	Socket CAN Configuration.....	3
1.2.	CAN2.0/FD Communication	5

1. SocketCAN

Confirm that the following packages have been installed:

- iproute2
- can-utils

1.1. Socket CAN Configuration

1). Check all SocketCAN devices details.

`$ip -details link show`

```
root@test-System-Product-Name:~# ip -details link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 minmtu 0 maxmtu 0 addrgenmode eui64 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 08:bf:b8:a4:9f:35 brd ff:ff:ff:ff:ff:ff promiscuity 0 minmtu 68 maxmtu 9216 addrgenmode none numtxqueues 4 numrxqueues 4 gso_max_size 65536 gso_max_segs 65535 parentbus pci parentdev 0000:05:00:0
    altname enp5s0
3: can0: <NOARP,ECHO> mtu 16 qdisc noop state DOWN mode DEFAULT group default qlen 10
    link/can promiscuity 0 minmtu 0 maxmtu 0
    can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
    can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
    can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
    termination 120 [ 0, 120 ]
    clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.4.auto
4: can1: <NOARP,ECHO> mtu 16 qdisc noop state DOWN mode DEFAULT group default qlen 10
    link/can promiscuity 0 minmtu 0 maxmtu 0
    can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
    can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
    can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
    termination 120 [ 0, 120 ]
    clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.5.auto
5: can2: <NOARP,ECHO> mtu 16 qdisc noop state DOWN mode DEFAULT group default qlen 10
    link/can promiscuity 0 minmtu 0 maxmtu 0
    can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
    can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
    can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
    termination 120 [ 0, 120 ]
    clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.6.auto
6: can3: <NOARP,ECHO> mtu 16 qdisc noop state DOWN mode DEFAULT group default qlen 10
    link/can promiscuity 0 minmtu 0 maxmtu 0
    can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
    can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
    can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
    termination 120 [ 0, 120 ]
    clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.7.auto
7: wlp0s20f3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DORMANT group default qlen 1000
    link/ether 30:05:05:89:48:92 brd ff:ff:ff:ff:ff:ff promiscuity 0 minmtu 256 maxmtu 2304 addrgenmode none numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus pci parentdev 0000:00:14:3
root@test-System-Product-Name:~#
```

SUNIX CAN Channel

2). Check single channel details, at this time, bus state should be 'STOPPED'.

`$ip -details link show [device]`

```
root@test-System-Product-Name:~# ip -details link show can0
3: can0: <NOARP,ECHO> mtu 16 qdisc noop state DOWN mode DEFAULT group default qlen 10
    link/can promiscuity 0 minmtu 0 maxmtu 0
    can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
    can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
    can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
    termination 120 [ 0, 120 ]
    clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.4.auto
```

3). Set CAN 2.0 or Set CAN FD.

🌈 CAN2.0: Set can channel up and set arbitration bitrate.

\$ip link set [channel] up type can bitrate [arbitration bitrate]

```
root@test-System-Product-Name:~# ip link set can0 up type can bitrate 1000000
```

➤ CAN2.0 arbitration bitrate ≤ 1Mbps

✚ CAN FD: Set can channel up, set arbitration bitrate and data bitrate, set FD on.

\$ip link set [channel] up type can bitrate [arbitration bitrate] dbitrte [data bitrate] fd on

```
root@test-System-Product-Name:~# ip link set can1 up type can bitrate 1000000 dbitrte 8000000 fd on
```

➤ 8Mbps ≥ data bitrate > arbitration bitrate

- 4). Make sure the bus status is ERROR-ACTIVE and bitrate configuration is correct.

\$ip -details link show

```
root@test-System-Product-Name:~# ip -details link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 minmtu 0 maxmtu 0 addrgenmode eui64 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
   link/ether 08:b7:b0:a4:9f:35 brd ff:ff:ff:ff:ff:ff promiscuity 0 minmtu 68 maxmtu 9216 addrgenmode none numtxqueues 4 numrxqueues 4 gso_max_size 65536 gso_max_segs 65535 parentbus pci parentdev 0000:05:00.0
   altname enp5s0
3: can0: <NOARP,UP,LOWER_UP,ECHO> mtu 16 qdisc pfifo_fast state UP mode DEFAULT group default qlen 10
   link/can promiscuity 0 minmtu 0 maxmtu 0
   can state ERROR-ACTIVE (berr-counter tx 0 rx 0) restart-ms 0
   bitrate 1000000 sample-point 0.750
   tq 12 prop-seg 29 phase-seg1 30 phase-seg2 20 sjw 10
   can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
   can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
   termination 120 [ 0, 120 ]
   clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.4.auto
4: can1: <NOARP,UP,LOWER_UP,ECHO> mtu 72 qdisc pfifo_fast state UP mode DEFAULT group default qlen 10
   link/can promiscuity 0 minmtu 0 maxmtu 0
   can <FD> state ERROR-ACTIVE (berr-counter tx 0 rx 0) restart-ms 0
   bitrate 1000000 sample-point 0.750
   tq 12 prop-seg 29 phase-seg1 30 phase-seg2 20 sjw 10
   can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
   dbitrte 8000000 dsample-point 0.700
   dtq 12 dprop-seg 3 dphase-seg1 3 dphase-seg2 3 dsjw 1
   can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
   termination 120 [ 0, 120 ]
   clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.5.auto
5: can2: <NOARP,ECHO> mtu 16 qdisc noop state DOWN mode DEFAULT group default qlen 10
   link/can promiscuity 0 minmtu 0 maxmtu 0
   can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
   can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
   can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
   termination 120 [ 0, 120 ]
   clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.6.auto
6: can3: <NOARP,ECHO> mtu 16 qdisc noop state DOWN mode DEFAULT group default qlen 10
   link/can promiscuity 0 minmtu 0 maxmtu 0
   can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
   can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
   can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
   termination 120 [ 0, 120 ]
   clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.7.auto
7: wlp0s20f3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DORMANT group default qlen 1000
   link/ether 30:05:05:89:48:92 brd ff:ff:ff:ff:ff:ff promiscuity 0 minmtu 256 maxmtu 2304 addrgenmode none numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus pci parentdev 0000:00:14.3
```

- 5). Set can channel down

\$ip link set [channel] down

```
root@test-System-Product-Name:~# ip link set can0 down
```

6). Make sure the bus status is STOPPED.

\$ip –details link show

```
3: can0: <NOARP,ECHO> mtu 16 qdisc pfifo_fast state DOWN mode DEFAULT group default qlen 10
link/can promiscuity 0 minmtu 0 maxmtu 0
can state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
  bitrate 1000000 sample-point 0.750
  tq 12 prop-seg 29 phase-seg1 30 phase-seg2 20 sjw 10
  can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
  can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
  termination 120 [ 0, 120 ]
  clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.4.auto
4: can1: <NOARP,ECHO> mtu 72 qdisc pfifo_fast state DOWN mode DEFAULT group default qlen 10
link/can promiscuity 0 minmtu 0 maxmtu 0
can <FD> state STOPPED (berr-counter tx 0 rx 0) restart-ms 0
  bitrate 1000000 sample-point 0.750
  tq 12 prop-seg 29 phase-seg1 30 phase-seg2 20 sjw 10
  can_sdc: tseg1 1..256 tseg2 1..256 sjw 1..256 brp 1..256 brp-inc 1
  dbitrates 8000000 dsample-point 0.700
  dtq 12 dprop-seg 3 dphase-seg1 3 dphase-seg2 3 dsjw 1
  can_sdc: dtseg1 1..64 dtseg2 1..32 dsjw 1..32 dbrp 1..256 dbrp-inc 1
  termination 120 [ 0, 120 ]
  clock 80000000 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 parentbus platform parentdev can_sdc.5.auto
```

1.2. CAN2.0/FD Communication

can-utils is a public can bus test utility, The document just introduce ‘cansend’ and ‘candump’, The two commands could test the basic data transmission.

About can-utils other commands, user could check the github:

<https://github.com/linux-can/can-utils>

Debian can-utils Package:

<https://packages.debian.org/en/sid/can-utils>

➤ cansend direction:

```
root@test-System-Product-Name:~# cansend
cansend - send CAN-frames via CAN_RAW sockets.

Usage: cansend <device> <can_frame>.

<can_frame>:
  <can_id>#{data}          for Classical CAN 2.0 data frames
  <can_id>#R{len}          for Classical CAN 2.0 data frames
  <can_id>#{data}_{dlc}     for Classical CAN 2.0 data frames
  <can_id>#R{len}_{dlc}    for Classical CAN 2.0 data frames
  <can_id>##<flags>{data}  for CAN FD frames

<can_id>:
  3 (SFF) or 8 (EFF) hex chars
{data}:
  0..8 (0..64 CAN FD) ASCII hex-values (optionally separated by '.')
{len}:
  an optional 0..8 value as RTR frames can contain a valid dlc field
_dlc:
  an optional 9..F data length code value when payload length is 8
<flags>:
  a single ASCII Hex value (0 .. F) which defines canfd_frame.flags

Examples:
5A1#11.2233.44556677.88 / 123#DEADBEEF / 5AA# / 123##1 / 213##311223344 /
1F334455#1122334455667788_B / 123#R / 00000123#R3 / 333#R8_E
```

➤ candump direction:

```
root@test-System-Product-Name:~# candump
candump - dump CAN bus traffic.

Usage: candump [options] <CAN interface>+
       (use CTRL-C to terminate candump)

Options:
  -t <type>      (timestamp: (a)bsolute/(d)elta/(z)ero/(A)bsolute w date)
  -H             (read hardware timestamps instead of system timestamps)
  -c             (increment color mode level)
  -i             (binary output - may exceed 80 chars/line)
  -a             (enable additional ASCII output)
  -S             (swap byte order in printed CAN data[] - marked with ' ' )
  -s <level>     (silent mode - 0: off (default) 1: animation 2: silent)
  -l             (log CAN-frames into file. Sets '-s 2' by default)
  -f <fname>     (log CAN-frames into file <fname>. Sets '-s 2' by default)
  -L             (use log file format on stdout)
  -n <count>     (terminate after reception of <count> CAN frames)
  -r <size>      (set socket receive buffer to <size>)
  -D             (Don't exit if a "detected" can device goes down)
  -d             (monitor dropped CAN frames)
  -e             (dump CAN error frames in human-readable format)
  -B             (display raw DLC values in {} for Classical CAN)
  -x             (print extra message infos, rx/tx brs esi)
  -T <msecs>     (terminate after <msecs> if no frames were received)

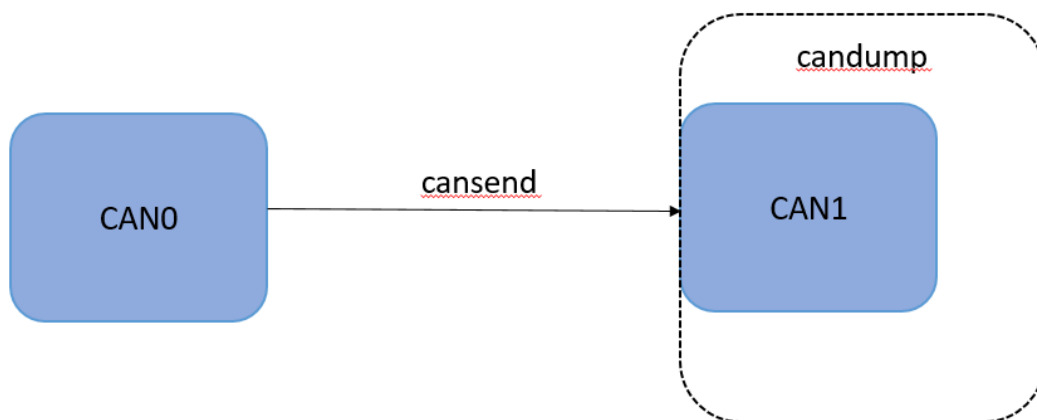
Up to 16 CAN interfaces with optional filter sets can be specified
on the commandline in the form: <ifname>[,filter]*

Filters:
  Comma separated filters can be specified for each given CAN interface:
  <can_id>:<can_mask>
    (matches when <received_can_id> & mask == can_id & mask)
  <can_id>~<can_mask>
    (matches when <received_can_id> & mask != can_id & mask)
  #<error_mask>
    (set error frame filter, see include/linux/can/error.h)
  [j|J]
    (join the given CAN filters - logical AND semantic)

CAN IDs, masks and data content are given and expected in hexadecimal values.
When the can_id is 8 digits long the CAN_EFF_FLAG is set for 29 bit EFF format.
Without any given filter all data frames are received ('0:0' default filter).

Use interface name 'any' to receive from all CAN interfaces.
```

➤ Test Condition



CAN 2.0

- 1). can0 send can frame by cansend

\$cansend [channel] [SID#data]

```
root@test-System-Product-Name: ~
File Edit View Search Terminal Help
root@test-System-Product-Name:~# cansend can0 1f123345#1122334455667788
root@test-System-Product-Name:~#
```

➤ Pattern Parsing

1f123345 → SID, send ID

→ CAN2.0

1122334455667788 → Data

- 2). can1 waiting the can frame by candump

\$candump [channel]

➤ Received CAN frame

```
root@test-System-Product-Name:~# candump can1
can1 1F123345 [8] 11 22 33 44 55 66 77 88
```

If test is finish, enter “ctrl + c” to exit candump.

CAN FD

- 1). Send CAN FD frame by cansend

\$cansend [channel] [SID##{flag}{data}]

```
root@test-System-Product-Name:~# cansend can0 7ff##31234567890abcdef1234567890abcdef
```

flag	RX/TX	BRS	ESI
0	RX	--	--
1	RX	✓	--
2	RX	--	✓
3	RX	✓	✓

➤ Pattern Parsing

7ff → SID, send ID

→ FD

3 → flag

1234567890abcdef1234567890abcdef → Data

2). can1 waiting the can frame(add -x to show flag) by candump

```
$candump [channel] -c -x
```

```
root@test-System-Product-Name:~# candump can1 -c -x
can1 RX B E 7FF [16] 12 34 56 78 90 AB CD EF 12 34 56 78 90 AB CD EF
```

If test is finish, enter “ctrl + c” to exit candump.